Calculus

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The unit circle and angles of any magnitude**

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 18 cm. Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

2 **[Foundational]** • **Target: The unit circle and angles of any magnitude**

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 11. Substituting $x = 2$ into $y = 2x + 7$ gives $y = 2(2) + 7 = 11$.

3 **[Foundational]** • **Target: The unit circle and angles of any magnitude**

In ΔABC studying the unit circle and angles of any magnitude, $m\angle A = 37^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 84°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (37^\circ + 59^\circ) = 84^\circ$.

4 **[Foundational]** • **Target: The unit circle and angles of any magnitude**

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 24 cm. Multiplying the original length by the scale factor gives $6 \times 4 = 24$ cm.

5 **[Foundational]** • **Target: The unit circle and angles of any magnitude**

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 13. Substituting $x = 3$ into $y = 2x + 7$ gives $y = 2(3) + 7 = 13$.

6 **[Foundational]** • **Target: The unit circle and angles of any magnitude**

In ΔABC studying the unit circle and angles of any magnitude, $m\angle A = 37^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 77°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (37^\circ + 66^\circ) = 77^\circ$.

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7 [Foundational] • Target: The unit circle and angles of any magnitude

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 30 cm. Multiplying the original length by the scale factor gives $6 \times 5 = 30$ cm.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The unit circle and angles of any magnitude

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Final Answer: 15. Substituting $x = 4$ into $y = 2x + 7$ gives $y = 2(4) + 7 = 15$.

9 [Intermediate] • Target: Trigonometric identities

For the model $f(x) = 5x + 8$ in trigonometric identities, evaluate $f(2)$.

Final Answer: 18. Substitute $x = 2$: $f(2) = 5(2) + 8 = 18$.

10 [Intermediate] • Target: Trigonometric identities

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

Final Answer: $x = 1$ and $x = 4$. Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

11 [Intermediate] • Target: Trigonometric identities

A data set recorded for trigonometric identities consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

12 [Intermediate] • Target: Trigonometric identities

For the model $f(x) = 5x + 8$ in trigonometric identities, evaluate $f(3)$.

Final Answer: 23. Substitute $x = 3$: $f(3) = 5(3) + 8 = 23$.

13 [Intermediate] • Target: Trigonometric identities

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

Final Answer: $x = 2$ and $x = 5$. Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

14 [Intermediate] • Target: Trigonometric identities

For the model $f(x) = 5x + 8$ in trigonometric identities, evaluate $f(4)$.

Final Answer: 28. Substitute $x = 4$: $f(4) = 5(4) + 8 = 28$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Trigonometric identities • Cross-Lesson: 4.01 Statistics & Probability

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

Final Answer: $x = 3$ and $x = 6$. Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

16 [Challenging] • Target: Trigonometric identities • Cross-Lesson: 6.01 AI SL — Specific Topics

For the model $f(x) = 5x + 8$ in trigonometric identities, evaluate $f(5)$.

Final Answer: 33. Substitute $x = 5$: $f(5) = 5(5) + 8 = 33$.

17 [Challenging] • Target: Trigonometric identities • Cross-Lesson: 1.01 Number & Algebra

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

Final Answer: $x = 4$ and $x = 7$. Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

18 [Challenging] • Target: Solving trigonometric equations 1 • Cross-Lesson: 2.01 Functions

For the model $f(x) = 6x + 9$ in solving trigonometric equations 1, evaluate $f(2)$.

Final Answer: 21. Substitute $x = 2$: $f(2) = 6(2) + 9 = 21$.

19 [Challenging] • Target: Solving trigonometric equations 1 • Cross-Lesson: 3.01 Geometry & Trigonometry

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

Final Answer: $x = 1$ and $x = 5$. Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

20 [Challenging] • Target: Solving trigonometric equations 1 • Cross-Lesson: 4.01 Statistics & Probability

A data set recorded for solving trigonometric equations 1 consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Exact values in right-angled triangles

In ΔABC studying exact values in right-angled triangles, $m\angle A = 35^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 100° ✓

b. 110°

c. 92°

d. 145°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 45^\circ) = 100^\circ$.

22 [Foundational] • Target: Exact values in right-angled triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?

a. 9 cm

b. 12 cm ✓

c. 16 cm

d. 36 cm

Multiplying the original length by the scale factor gives $4 \times 3 = 12$ cm.

23 [Foundational] • Target: Exact values in right-angled triangles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$?

a. 13

b. 19

c. 17 ✓

d. 21

Substituting $x = 3$ into $y = 4x + 5$ gives $y = 4(3) + 5 = 17$.

24 [Foundational] • Target: Exact values in right-angled triangles

In ΔABC studying exact values in right-angled triangles, $m\angle A = 35^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 79° ✓

b. 89°

c. 71°

d. 145°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 66^\circ) = 79^\circ$.

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25 [Foundational] • Target: Sine rule and the ambiguous case

For the model $f(x) = 3x + 6$ in sine rule and the ambiguous case, evaluate $f(2)$.

a. 12 ✓

b. 15

c. 6

d. 16

Substitute $x = 2$: $f(2) = 3(2) + 6 = 12$.

26 [Foundational] • Target: Sine rule and the ambiguous case

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ **and** $x = 7$ ✓

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

27 [Foundational] • Target: Sine rule and the ambiguous case

For the model $f(x) = 3x + 6$ in sine rule and the ambiguous case, evaluate $f(5)$.

a. 21 ✓

b. 24

c. 15

d. 25

Substitute $x = 5$: $f(5) = 3(5) + 6 = 21$.

28 [Foundational] • Target: The unit circle and angles of any magnitude

In ΔABC studying the unit circle and angles of any magnitude, $m\angle A = 37^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 98° ✓

b. 108°

c. 90°

d. 143°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (37^\circ + 45^\circ) = 98^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Exact values in right-angled triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm ✓

c. 12 cm

d. 16 cm

Multiplying the original length by the scale factor gives $4 \times 2 = 8$ cm.

30 [Intermediate] • Target: Exact values in right-angled triangles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$?

a. 5

b. 11

c. 9 ✓

d. 13

Substituting $x = 1$ into $y = 4x + 5$ gives $y = 4(1) + 5 = 9$.

31 [Intermediate] • Target: Exact values in right-angled triangles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$?

a. 9

b. 15

c. 13 ✓

d. 17

Substituting $x = 2$ into $y = 4x + 5$ gives $y = 4(2) + 5 = 13$.

32 [Intermediate] • Target: Exact values in right-angled triangles

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm ✓

c. 20 cm

d. 64 cm

Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

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33 [Intermediate] • Target: Exact values in right-angled triangles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$?

a. 17

b. 23

c. 21 ✓

d. 25

Substituting $x = 4$ into $y = 4x + 5$ gives $y = 4(4) + 5 = 21$.

34 [Intermediate] • Target: Sine rule and the ambiguous case

A data set recorded for sine rule and the ambiguous case consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

35 [Intermediate] • Target: Sine rule and the ambiguous case

For the model $f(x) = 3x + 6$ in sine rule and the ambiguous case, evaluate $f(3)$.

a. 15 ✓

b. 18

c. 9

d. 19

Substitute $x = 3$: $f(3) = 3(3) + 6 = 15$.

36 [Intermediate] • Target: Sine rule and the ambiguous case

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

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37 [Intermediate] • Target: The unit circle and angles of any magnitude

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm ✓

c. 18 cm

d. 24 cm

Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The unit circle and angles of any magnitude • Cross-Lesson: 2.01 Functions

In ΔABC studying the unit circle and angles of any magnitude, $m\angle A = 37^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 91° ✓

b. 101°

c. 83°

d. 143°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (37^\circ + 52^\circ) = 91^\circ$.

39 [Challenging] • Target: The unit circle and angles of any magnitude • Cross-Lesson: 3.01 Geometry & Trigonometry

In ΔABC studying the unit circle and angles of any magnitude, $m\angle A = 37^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 84° ✓

b. 94°

c. 76°

d. 143°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (37^\circ + 59^\circ) = 84^\circ$.

40 [Challenging] • Target: The unit circle and angles of any magnitude • Cross-Lesson: 4.01 Statistics & Probability

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm ✓

c. 36 cm

d. 150 cm

Multiplying the original length by the scale factor gives $6 \times 5 = 30$ cm.

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41 [Challenging] • Target: The unit circle and angles of any magnitude • Cross-Lesson: 6.01 AI SL — Specific Topics

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 13

b. 17

c. 15 ✓

d. 17

Substituting $x = 4$ into $y = 2x + 7$ gives $y = 2(4) + 7 = 15$.

42 [Challenging] • Target: Solving trigonometric equations 1 • Cross-Lesson: 1.01 Number & Algebra

A data set recorded for solving trigonometric equations 1 consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

43 [Challenging] • Target: Solving trigonometric equations 1 • Cross-Lesson: 2.01 Functions

For the model $f(x) = 6x + 9$ in solving trigonometric equations 1, evaluate $f(4)$.

a. 33 ✓

b. 39

c. 24

d. 37

Substitute $x = 4$: $f(4) = 6(4) + 9 = 33$.

44 [Challenging] • Target: Solving trigonometric equations 1 • Cross-Lesson: 3.01 Geometry & Trigonometry

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ **and** $x = 8$ ✓

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

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45 [Challenging] • Target: Solving trigonometric equations 2 • Cross-Lesson: 4.01 Statistics & Probability

For the model $f(x) = 2x + 3$ in solving trigonometric equations 2, evaluate $f(3)$.

a. 9 ✓

b. 11

c. 6

d. 13

Substitute $x = 3$: $f(3) = 2(3) + 3 = 9$.

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